

Classic Architectures

LeNet

LeNet

Digit recognition: 10 classes

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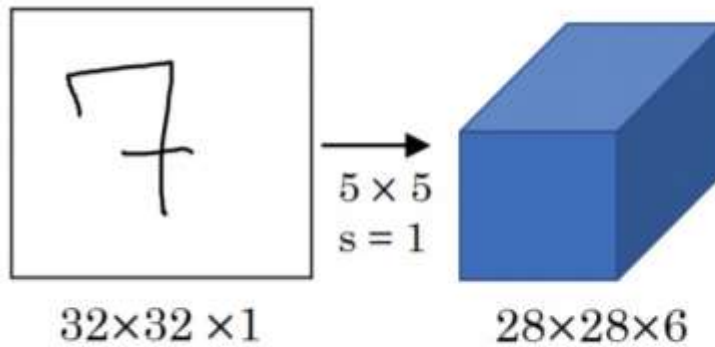


$32 \times 32 \times 1$

Input: 32×32
grayscale
images

LeNet

Digit recognition: 10 classes

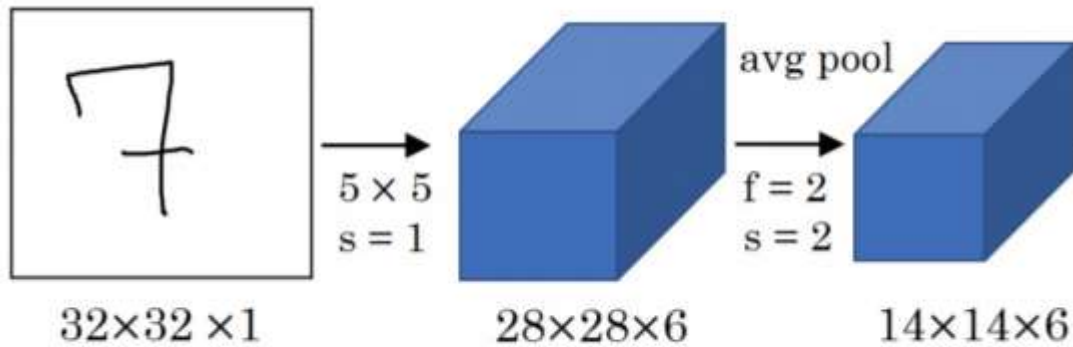


Input: 32×32
grayscale
images

Valid convolution:
size shrinks
6 conv filters in the
first layer

LeNet

Digit recognition: 10 classes

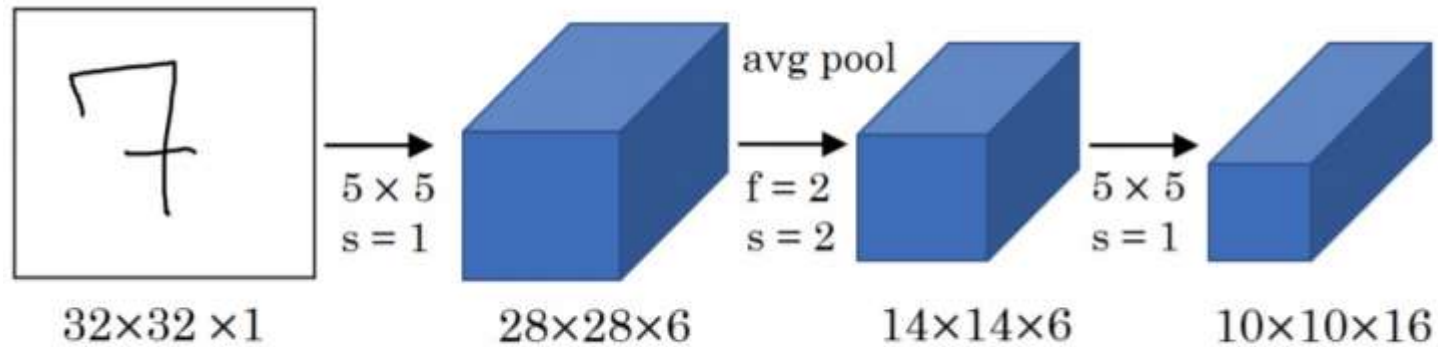


Input: 32×32
grayscale
images

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6 conv filters in the
first layer

LeNet

Digit recognition: 10 classes



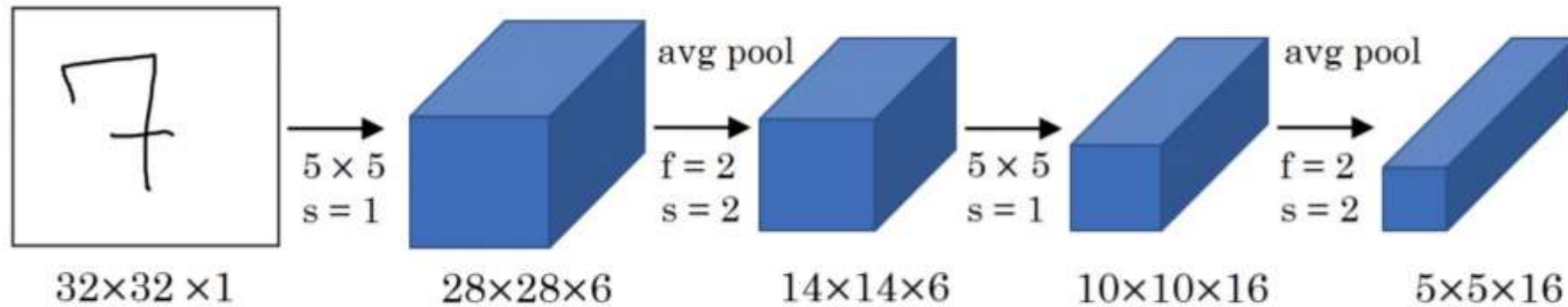
Input: 32×32
grayscale
images

Valid convolution:
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6 conv filters in the
first layer

Again valid
convolutions.

LeNet

Digit recognition: 10 classes



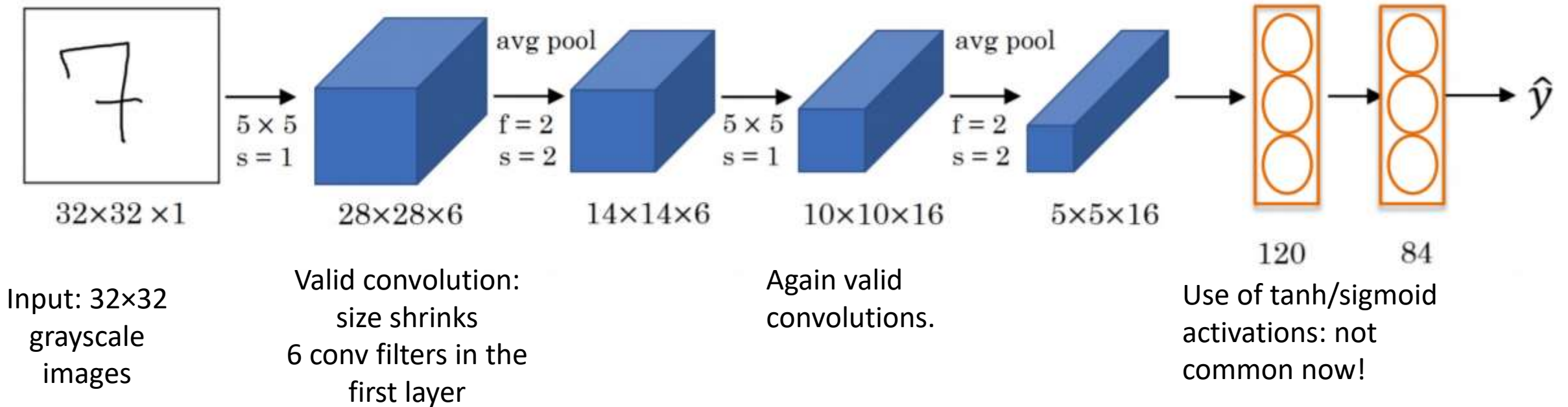
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LeNet

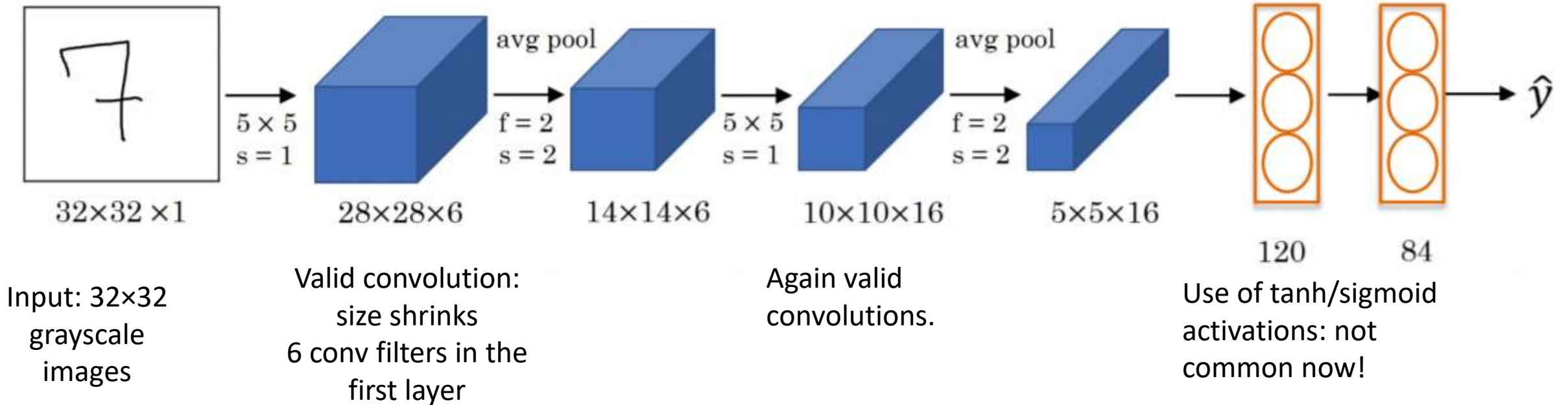
Digit recognition: 10 classes



LeNet

Digit recognition: 10 classes

Average pooling
was used, now
max pooling is
much more
common

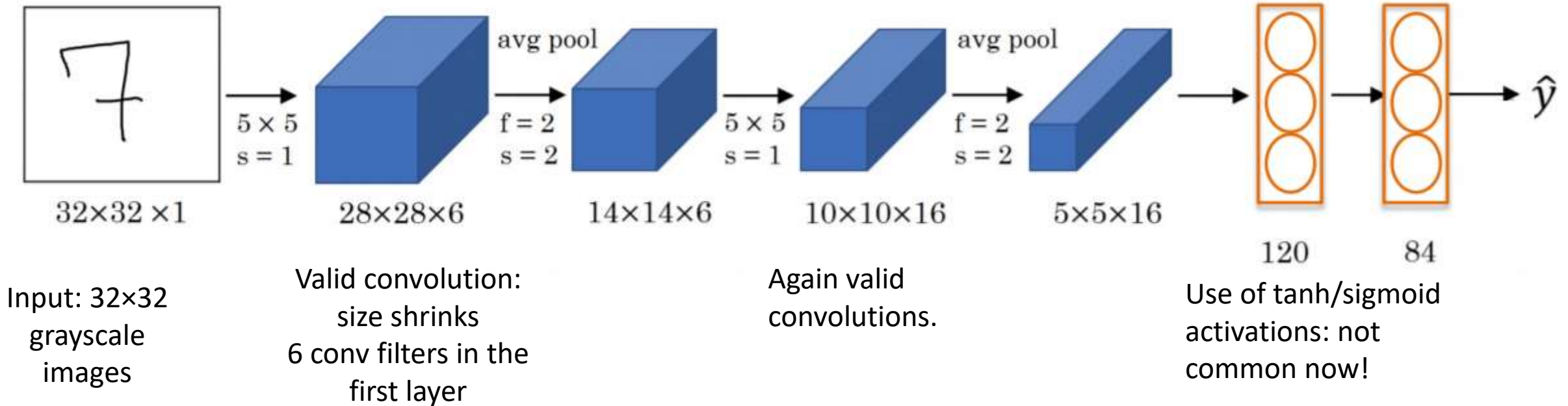


LeNet

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60K parameters

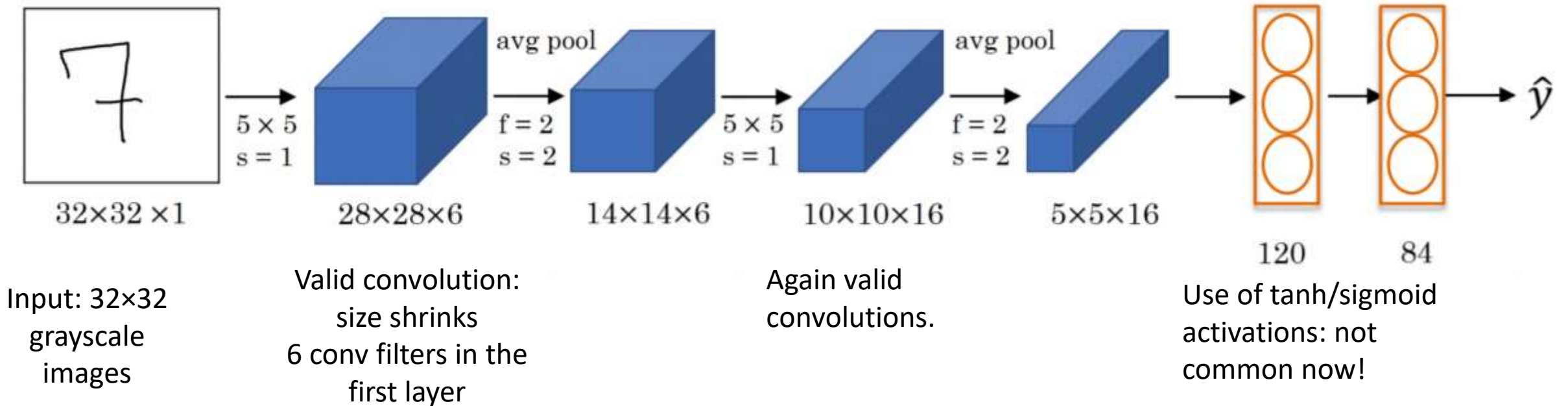


LeNet

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60K parameters



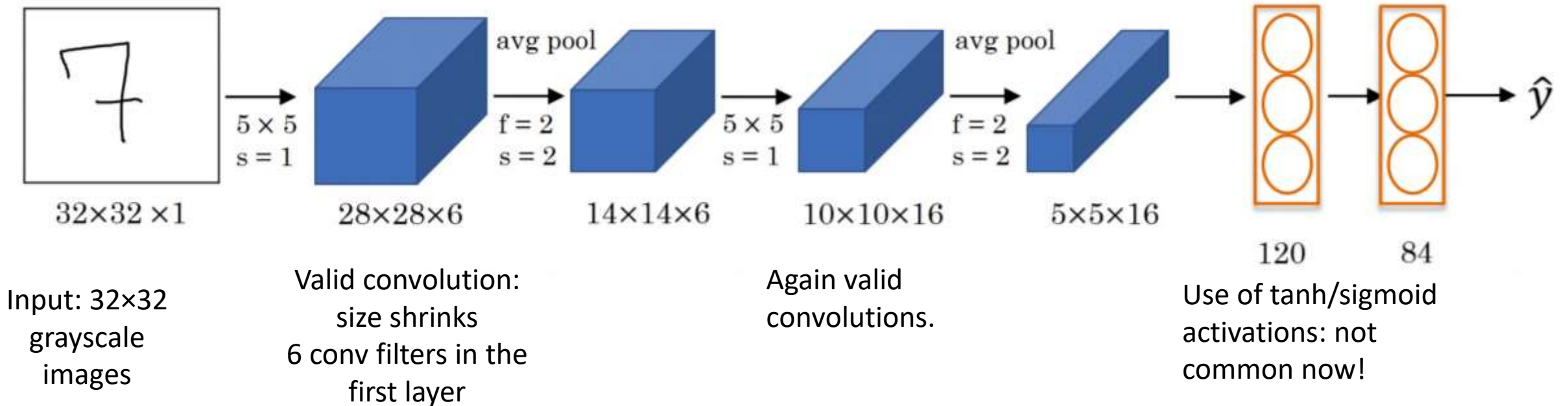
Conv -> Pool -> Conv -> Pool -> Conv -> FC

LeNet

Digit recognition: 10 classes

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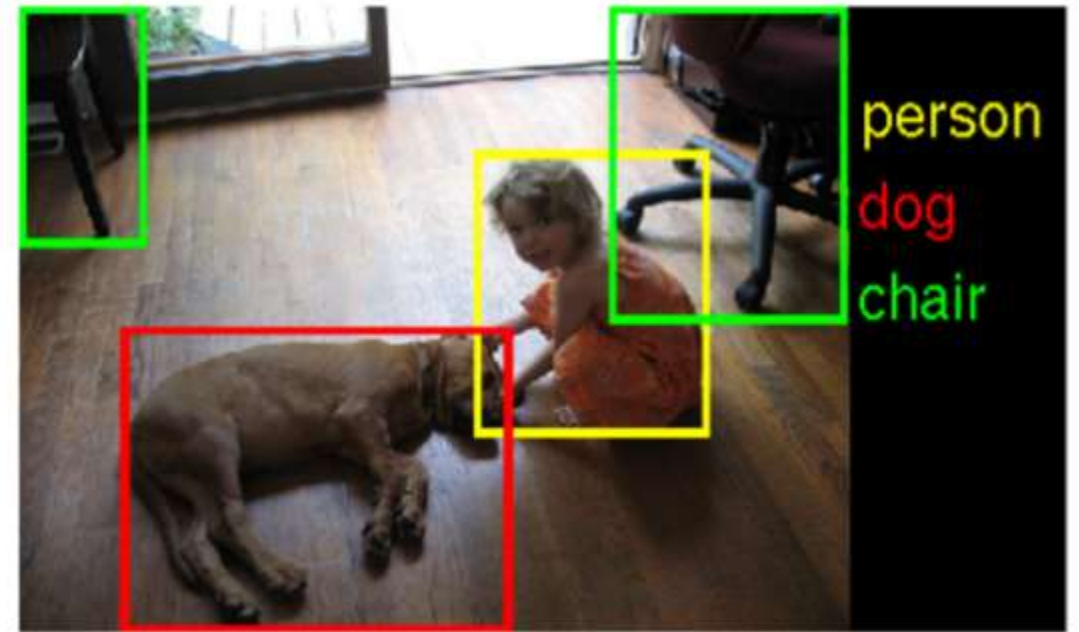
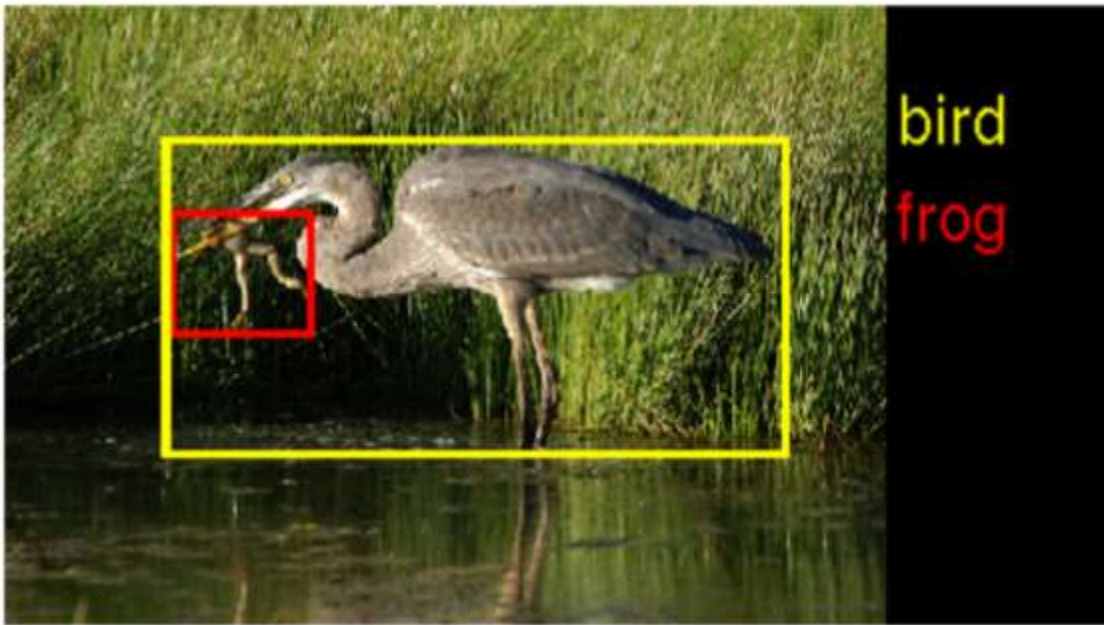
60K parameters



Conv -> Pool -> Conv -> Pool -> Conv -> FC

As we go deeper: Width, Height ↓ Number of Filters ↑

ImageNet Dataset: ImageNet Large Scale Visual Recognition Competition (ILSVRC)

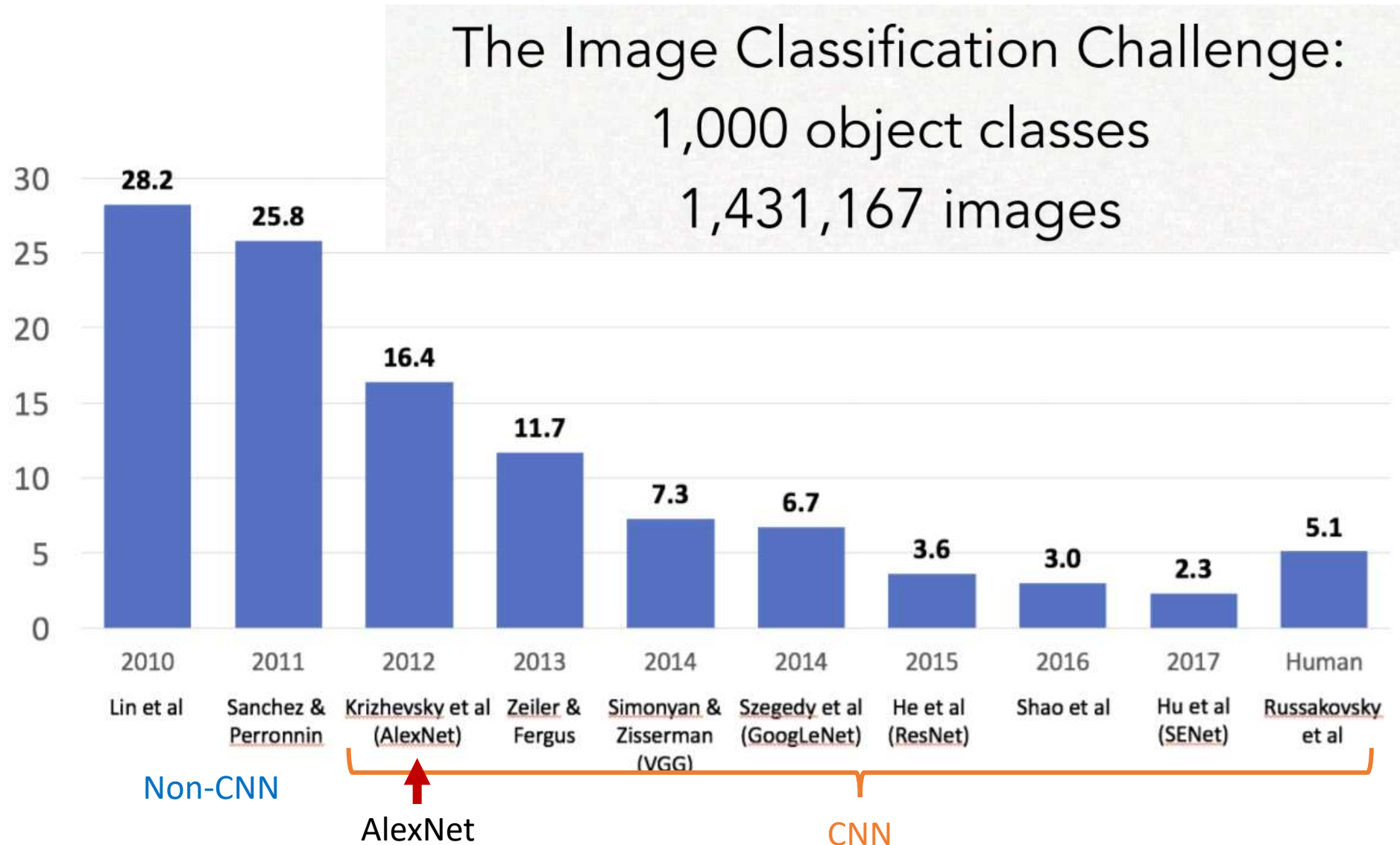


[Russakovsky et al., IJCV'15] "ImageNet Large Scale Visual Recognition Challenge."

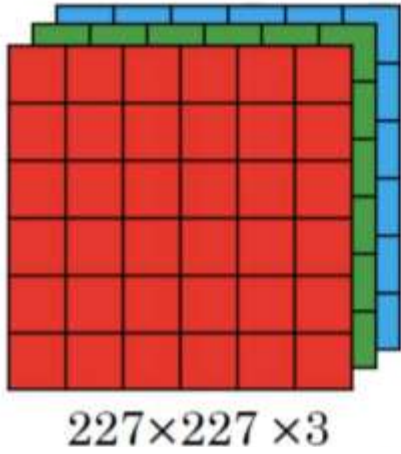
Common Performance Metrics

- **Top-1 score:** check if a sample's top class (i.e. the one with highest probability) is the same as its target label
- **Top-5 score:** check if your label is in your 5 first predictions (i.e. predictions with 5 highest probabilities)
- **Top-5 error:** percentage of test samples for which the correct class was not in the top 5 predicted classes

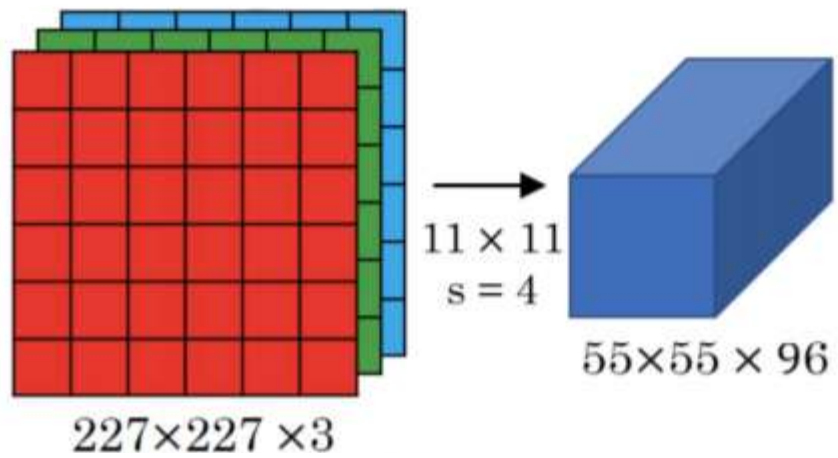
AlexNet



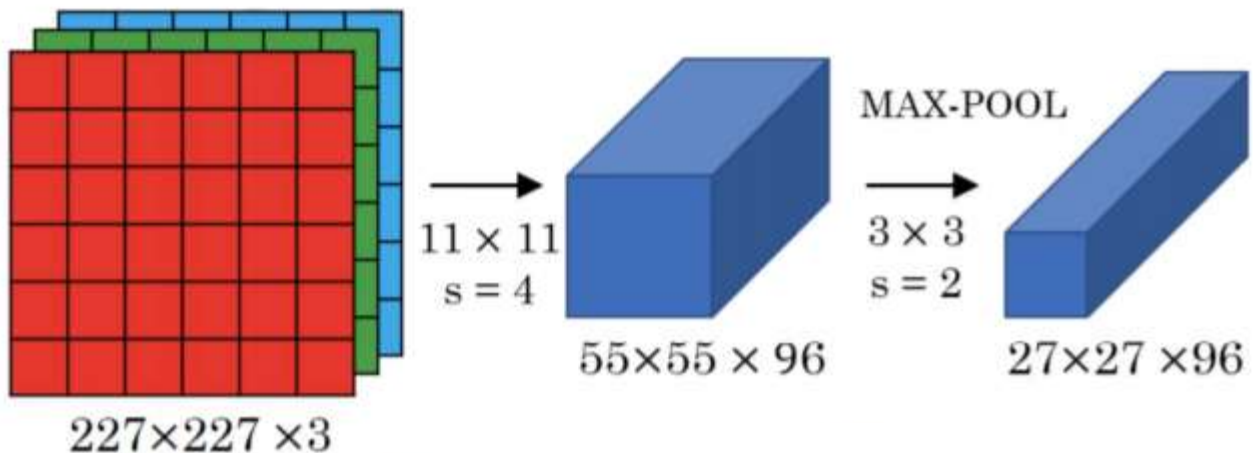
AlexNet



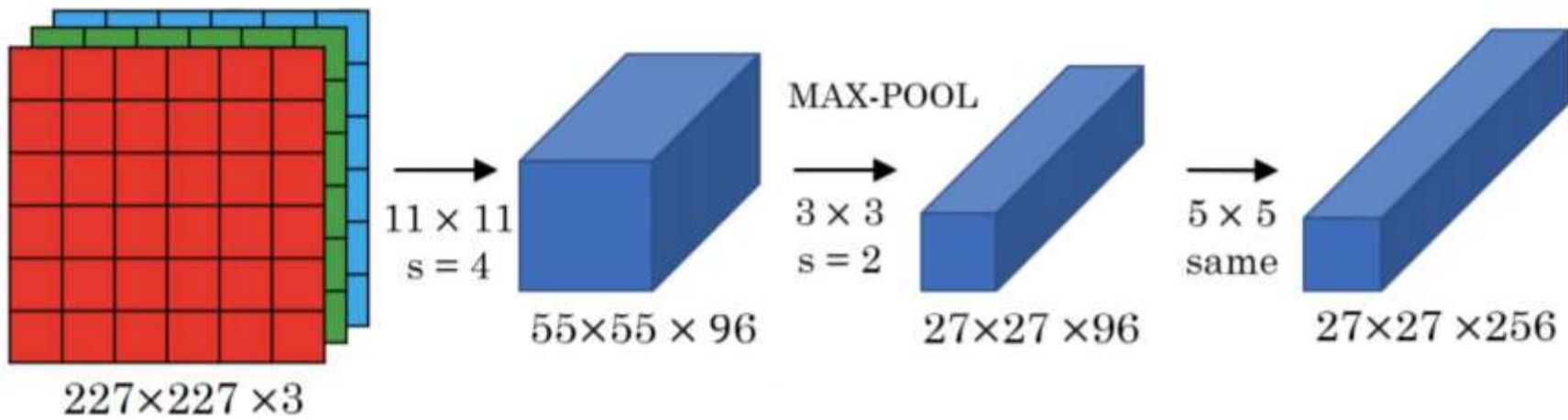
AlexNet



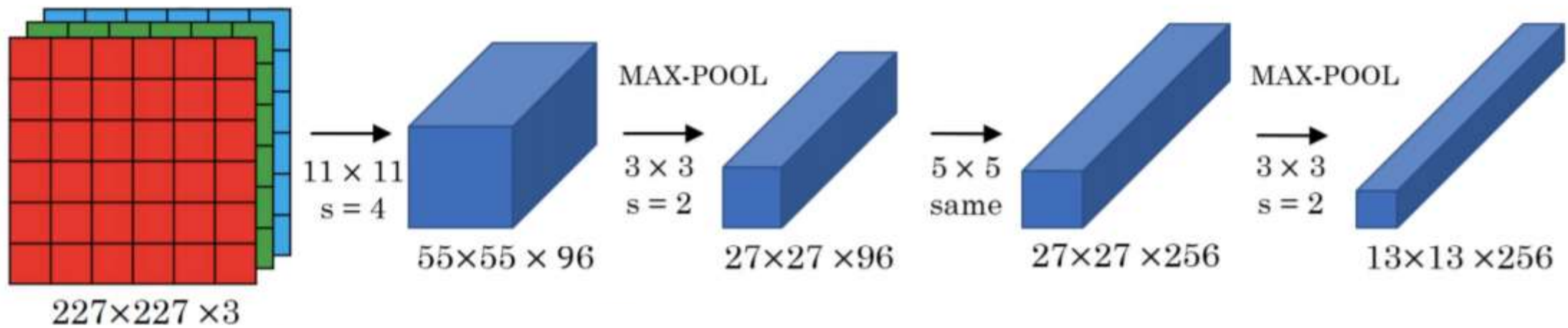
AlexNet



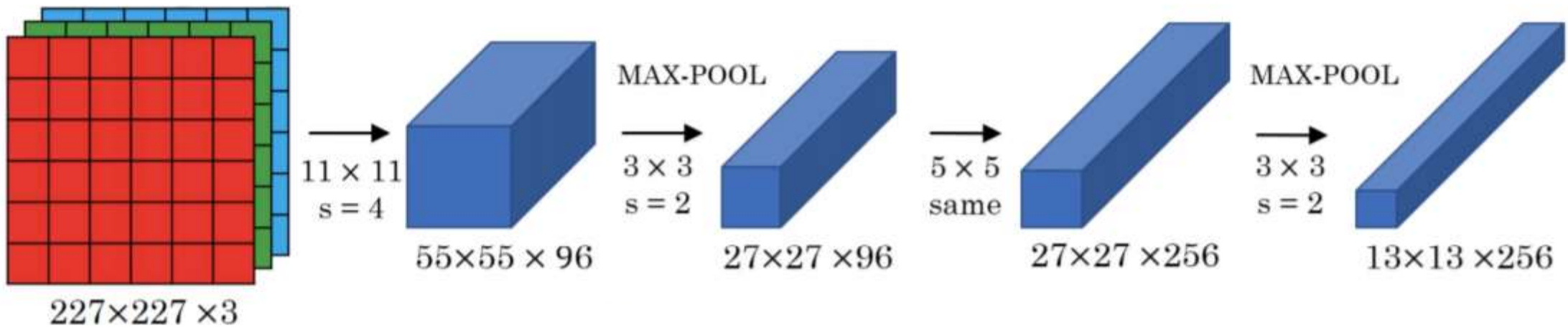
AlexNet



AlexNet

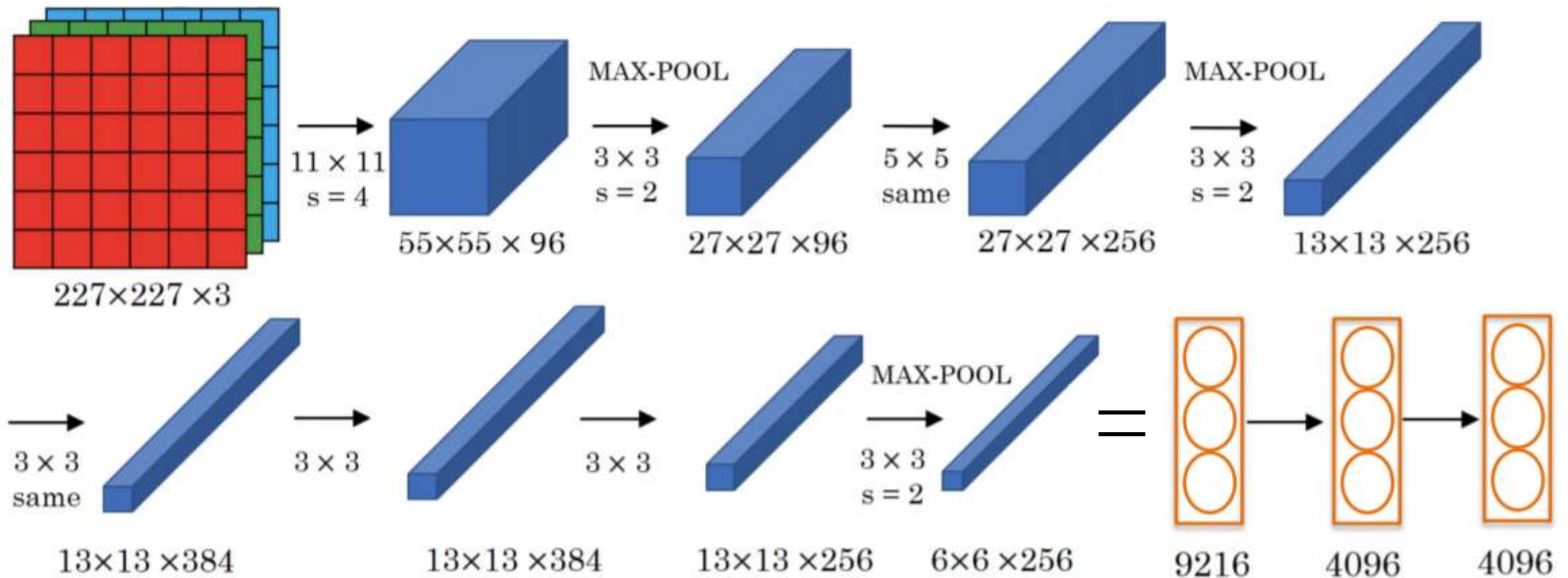


AlexNet



As with LeNet: Width, Height ↓ Number of Filters ↑

AlexNet



Softmax for 1000 classes

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

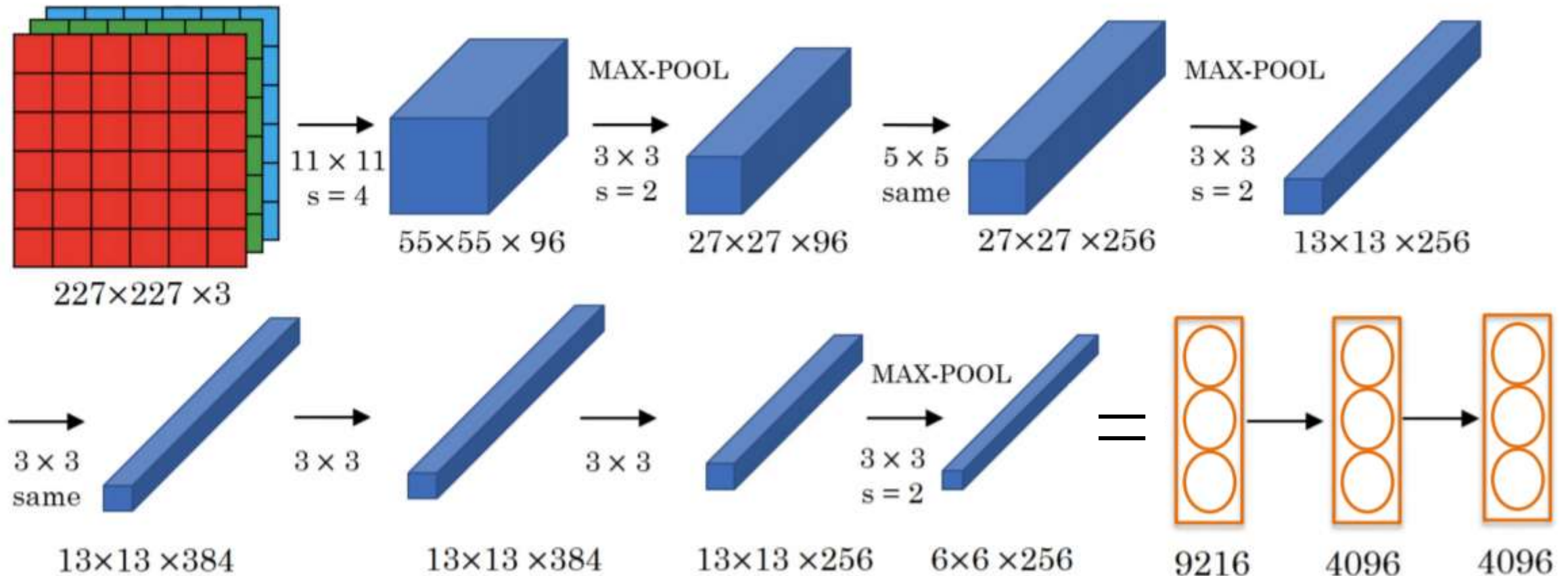
[Krizhevsky et al. NIPS'12] AlexNet

AlexNet

Similar to LeNet but much bigger (~ 1000 times)

60M parameters

Use of ReLU instead of tanh/sigmoid



Softmax for 1000 classes

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

[Krizhevsky et al. NIPS'12] AlexNet

VGGNet

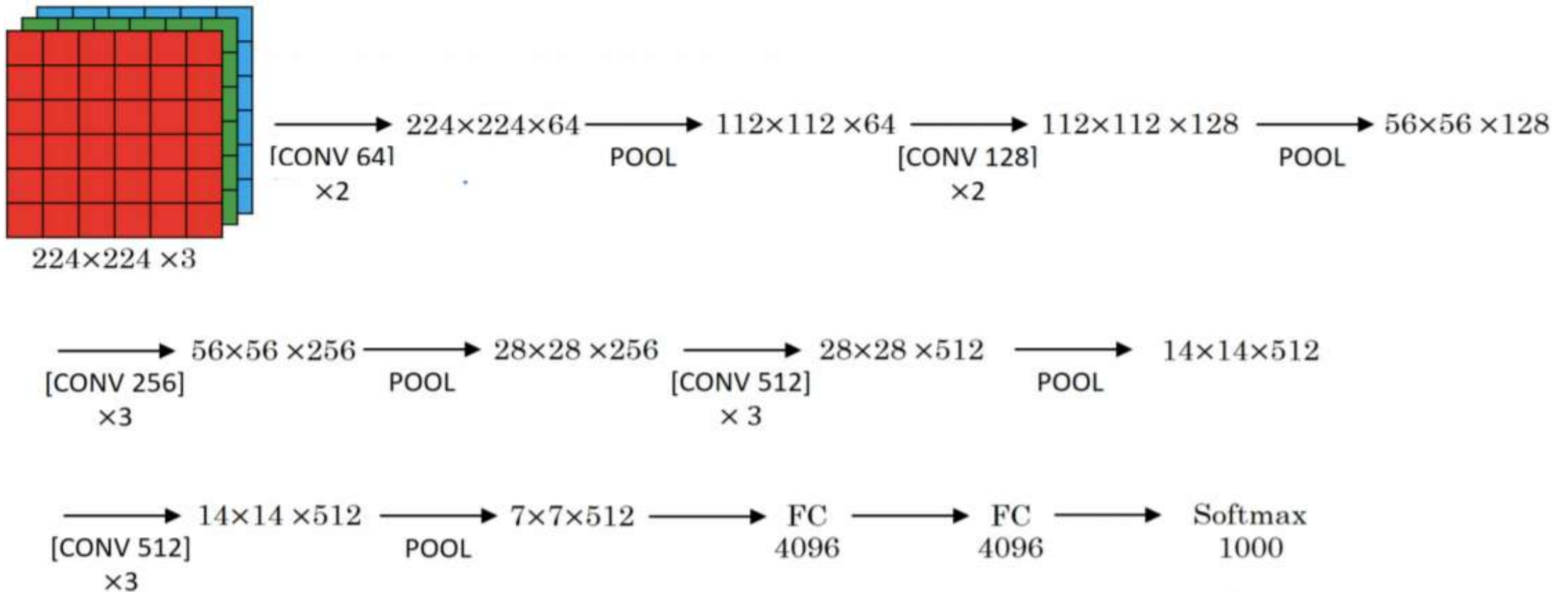
VGGNet

CONV = 3x3 filters with stride 1,
same convolutions

MAXPOOL = 2x2 filters with stride 2

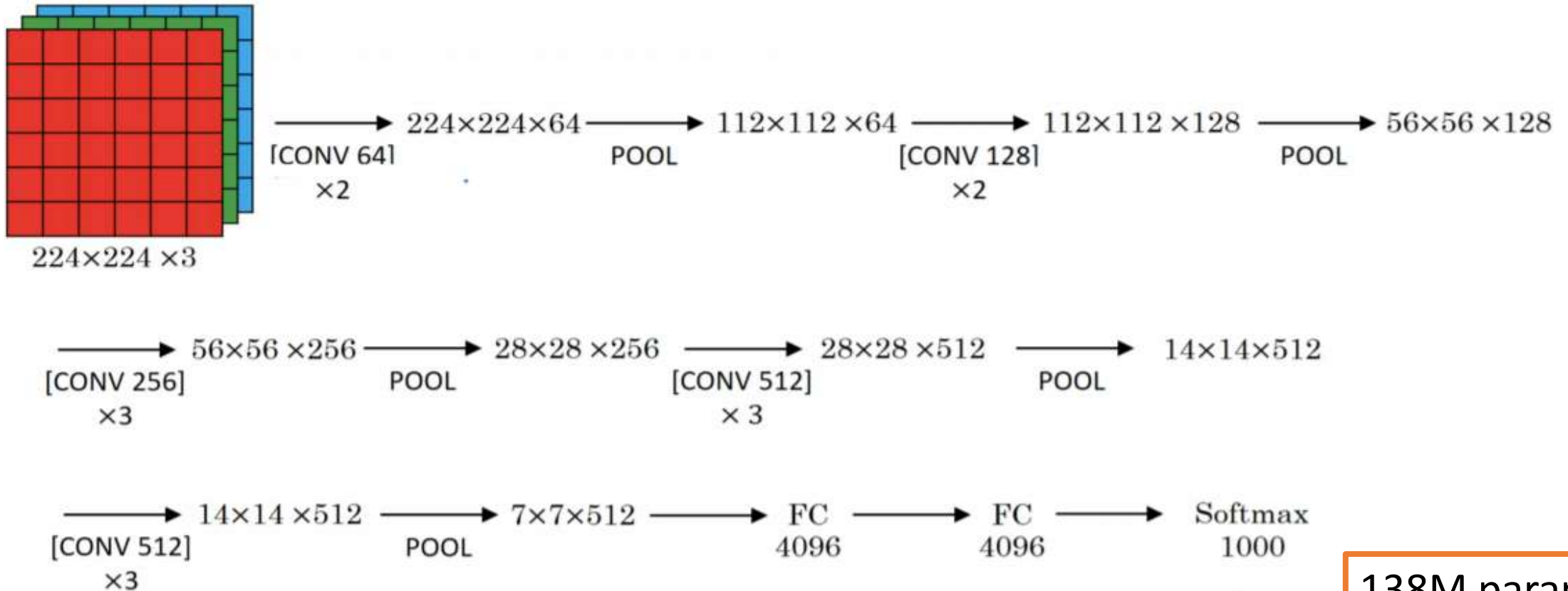
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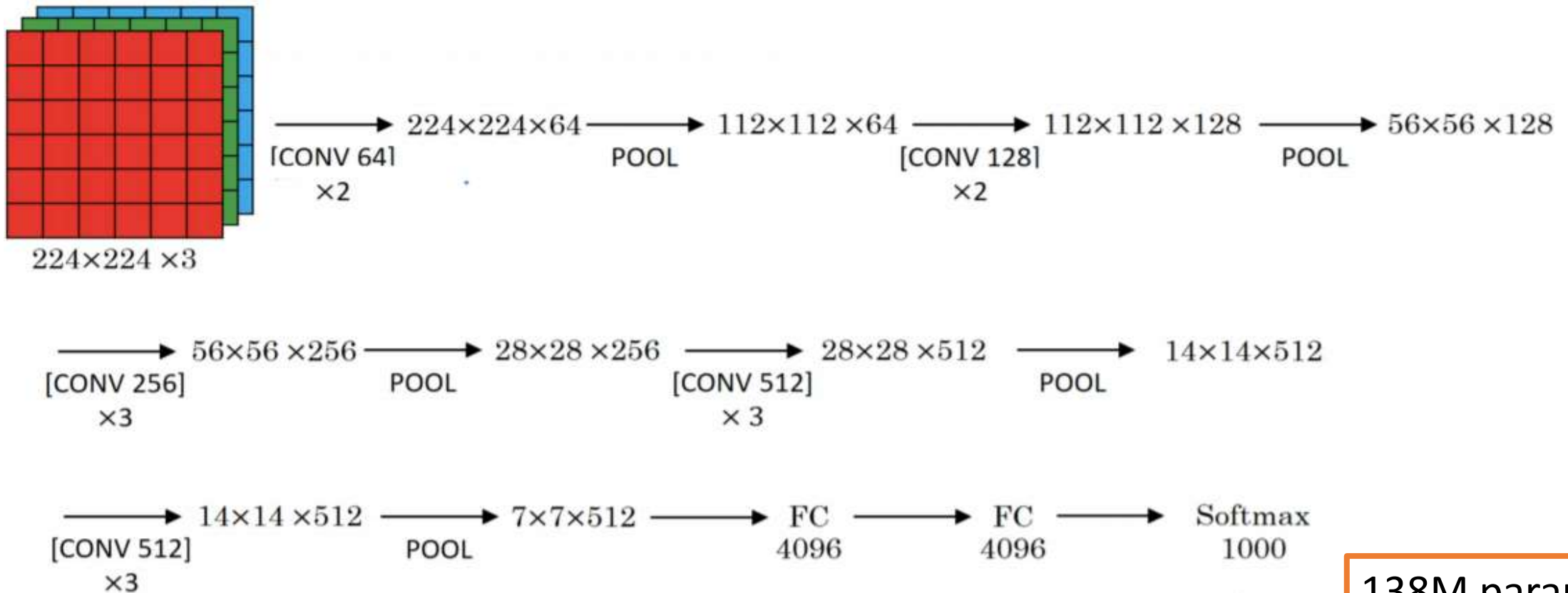
138M parameters

Conv -> Pool -> Conv -> Pool -> Conv -> FC

Called VGG-16: 16 layers that have weights

VGGNet

CONV = 3x3 filters with stride 1,
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MAXPOOL = 2x2 filters with stride 2



138M parameters

Conv -> Pool -> Conv -> Pool -> Conv -> FC

As we go deeper: Width, Height ↓ Number of Filters ↑

Called VGG-16: 16 layers that have weights

Large but simplicity makes it appealing

[Simonyan and Zisserman ICLR'15] VGGNet

VGGNet

Small filters, Deeper networks

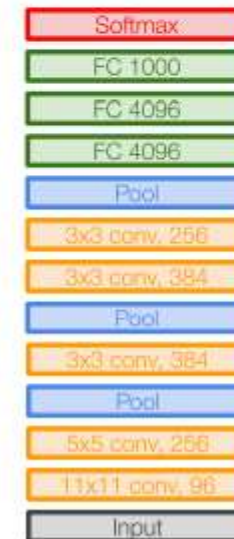
8 layers (AlexNet)

-> 16 - 19 layers (VGG16Net)

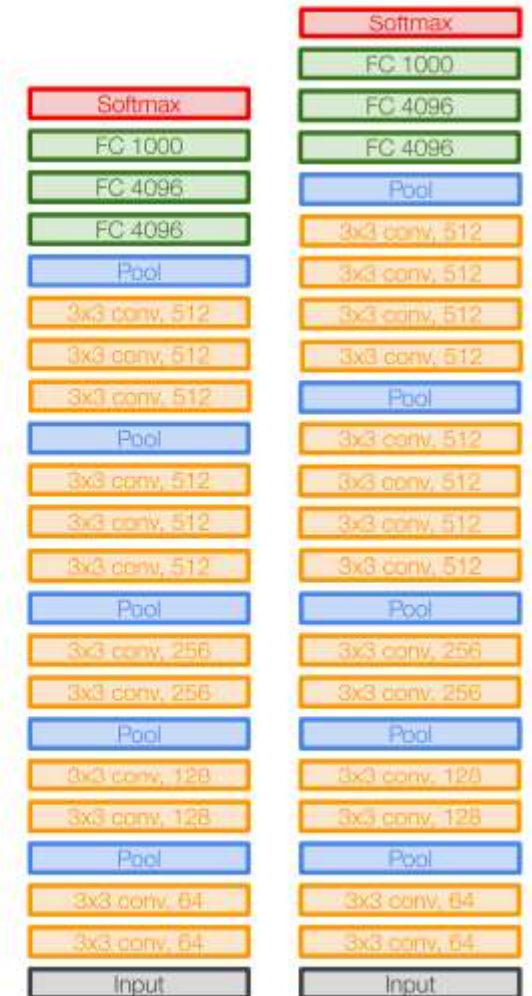
Only 3x3 CONV stride 1, pad 1
and 2x2 MAX POOL stride 2

11.7% top 5 error in ILSVRC'13 (ZFNet)

-> 7.3% top 5 error in ILSVRC'14



AlexNet

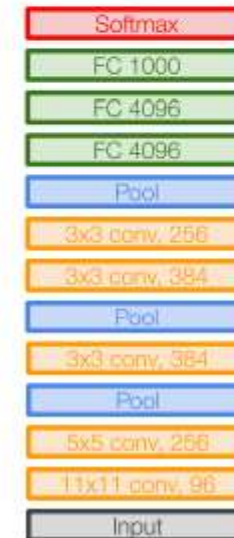


VGG16

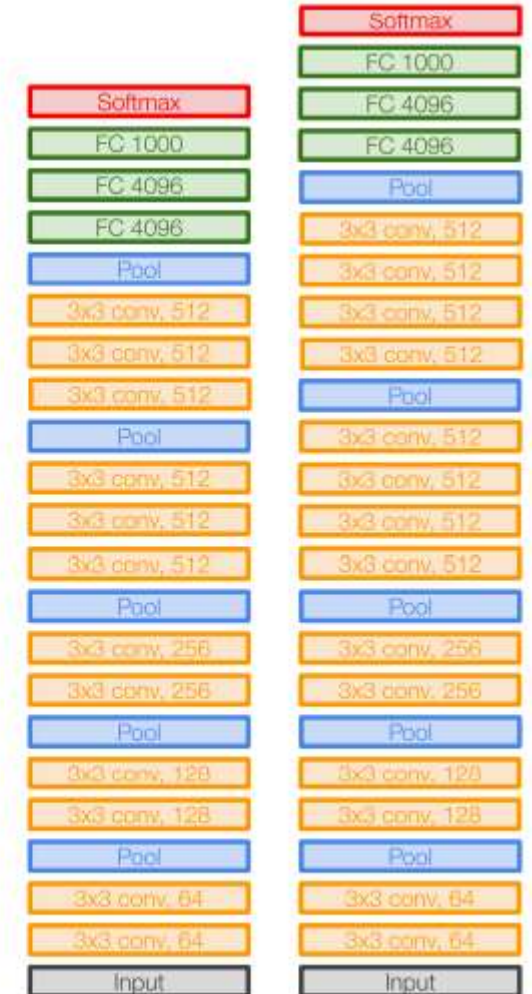
VGG19

VGGNet

Q: Why use smaller filters?



AlexNet



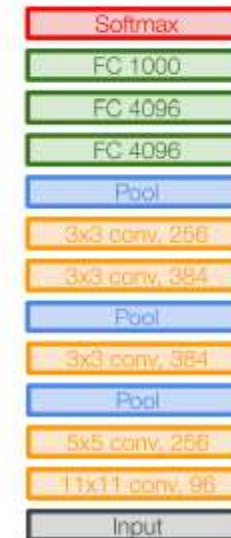
VGG16

VGG19

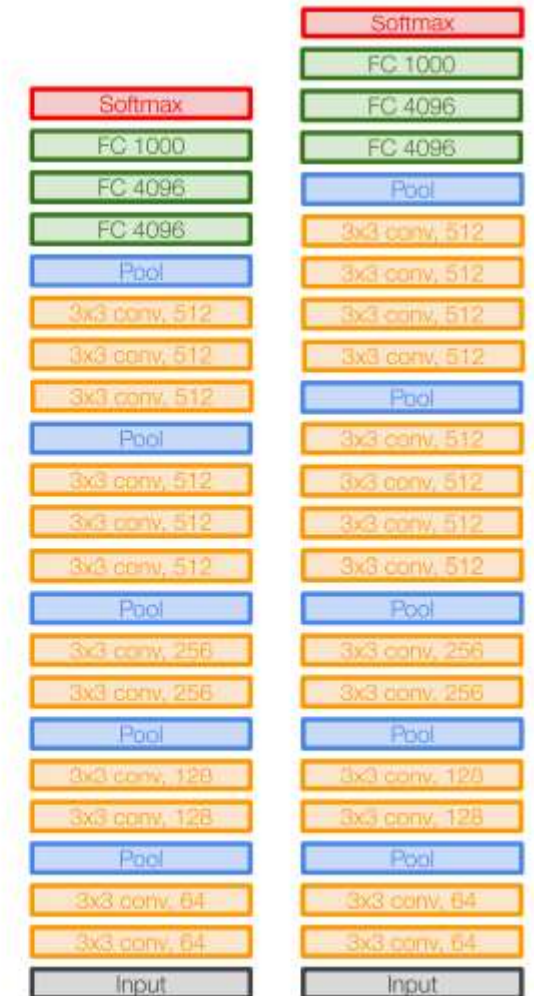
VGGNet

Q: Why use smaller filters?

Stack of three 3x3 conv (stride 1) layers has same effective receptive field as one 7x7 conv layer



AlexNet



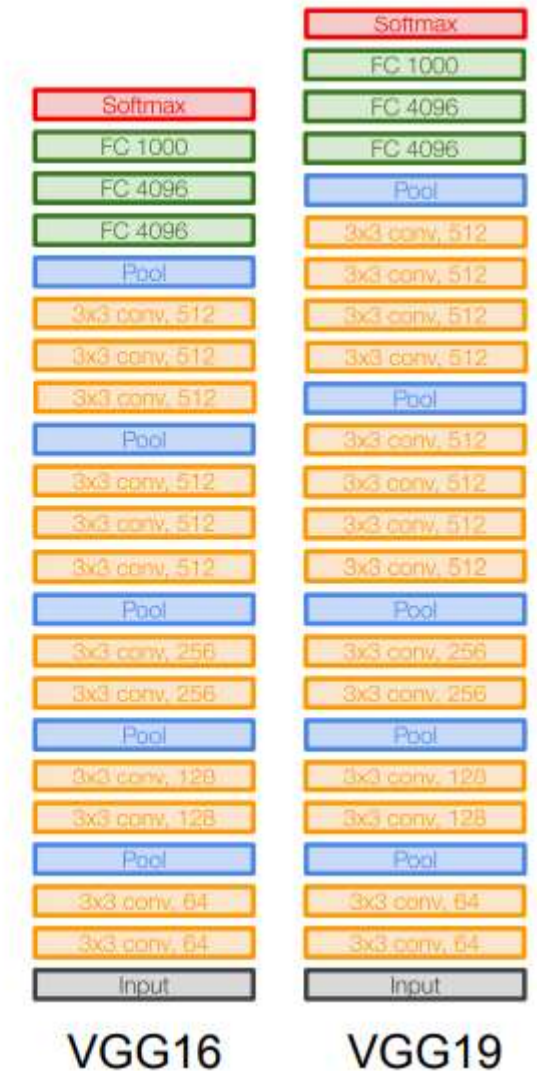
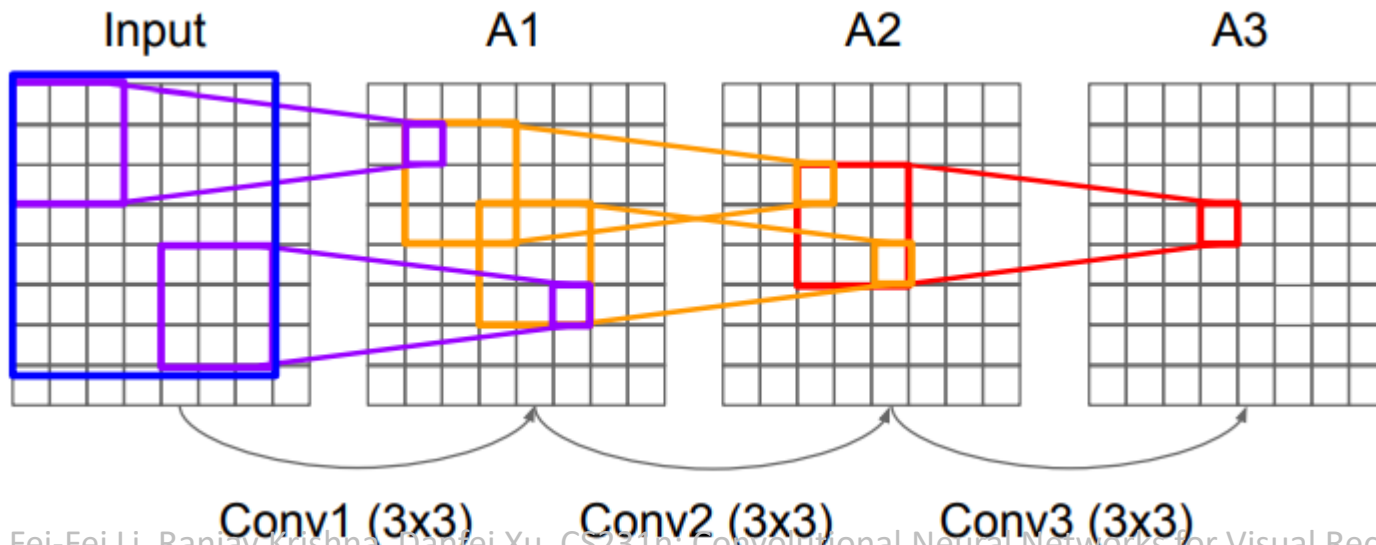
VGG16

VGG19

VGGNet

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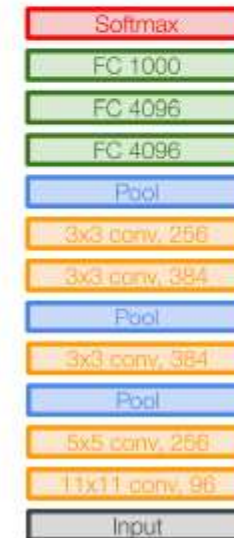


VGGNet

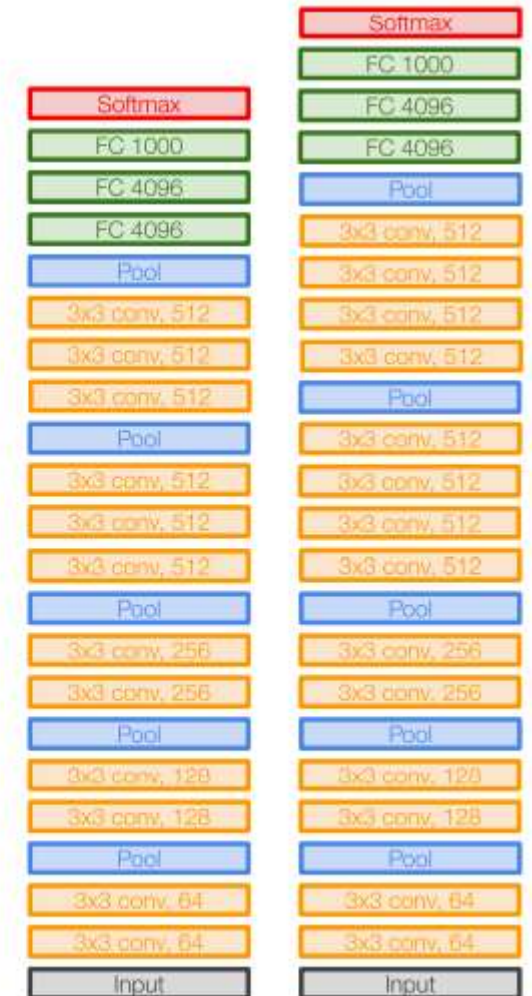
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But deeper, more non-linearities



AlexNet



VGG16

VGG19